

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electrical Engineering) SEMESTER - 5 Winter 2025 (Regular)

Course :Bachelor of Technology (Electrical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTEEC503 - POWER ELECTRONICS

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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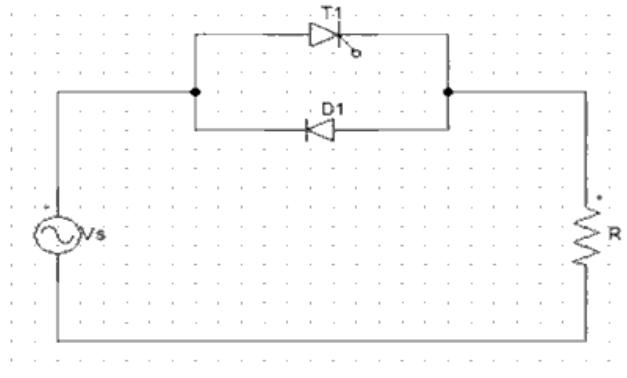
Q1. Objective type questions. (Compulsory Question) 12

- 1 Single phase one pulse diode rectifier is feeding an RL load with freewheeling diode across the load. For conduction angle β , the main diode & freewheeling diode will respectively conduct for
A. π , $\pi - \beta$ B. π , $\beta - \pi$ C. β , π D. $\beta - \pi$, π
- 2 When the Thyristor is reversed biased, the no of blocked p-n junction is
A. 1 B. 2 C. 3 D. 4
- 3 In 3 phase full converter, the six SCR's are fired at an interval of
A. 30° B. 60° C. 90° D. 120°
- 4 If a step-up chopper's switch is always kept OFF, what would be the ideal output voltage
A. 0 B. Infinity C. Equal to input voltage D. Dependent on load
- 5 In non-circulating current mode dual converters, the circulating current is avoided by
A. connecting a series reactor B. maintaining $\alpha_1 + \alpha_2 = 180^\circ$ C. operating only one converter D. adding an extra SCR
- 6 The control strategy in which on and off time is guided by the pervious set of values of a certain parameter is called as
A. time ratio control B. pulse width modulation
B. current limit control D. constant frequency scheme
- 7 The peak value of the line voltage in case of 120° mode of operation of a three-phase bridge inverter is
A. $V_s/2$ B. $3V_s/2$ C. $V_s/\sqrt{2}$ D. V_s
- 8 In current source inverters (CSI)
A. the amplitude of the output current is independent of the load
B. the amplitude of the output current depends on the load
C. the amplitude of the output voltage is independent of the load
D. None

- 9 In single-phase modulation of PWM inverters, the 5th order harmonic can be eliminated if the pulse width is made equal to _____

A. 120° B. 60° C. 72° D. 30°

- 10 In the below given voltage controller circuit



- A. the positive half cycle at the load is same as the supply V_s
 B. the negative half cycle at the load is same as the supply V_s
 C. the positive and negative half cycles at the load are identical to the supply
 D. None
- 11 A single-phase full converter B-2 type connection has a RLE type of motor load connected. The minimum requirement to turn-on the device is
- A. $\alpha > 30^\circ$ B. $V_m \sin \alpha > E$ C. $V_m \sin \alpha < E$ D. $\alpha < 30^\circ$
- 12 In _____ type of modulation method, the pulse width is not equal for all the pulses.
- A. multiple pulse width modulation B. single pulse width modulation
 C. sinusoidal pulse width modulation D. All of the above

Q2. Solve the following.

- A) Explain Switching characteristics of SCR with suitable waveforms 6
 B) A CE n-p-n transistor has current gain $\beta=40$. The load resistance $R_C = 10\Omega$, dc supply $V_{CC} = 130V$ and input voltage to base circuit, $V_B = 10V$. For $V_{CES} = 1.0V$ and $V_{BES} = 1.5V$, Calculate (a) The value of R_B for operation in saturated state. (b) The value of R_B for an overdrive factor 5 (c) Forced current gain 6

Q3. Solve the following.

- A) Explain single phase full wave controlled rectifier with RL load with circuit diagram, waveforms and necessary equations. 6
 B) Discuss the effect of source impedance on the performance of converter with waveforms & required equations. 6

Q4. Solve Any Two of the following.

- A) Explain buck converter with circuit diagram, equations and necessary waveforms 6
 B) A single phase full converter supplies power to RLE load. The source voltage is 230V, 50Hz and for load $R=2\Omega$, $L=10mH$, $E=100V$. For firing angle of 30° , find the average value of output current & output voltage for load current extinguishes at 200° 6
 C) Explain different types of control strategies of Chopper 6

- Q5. Solve Any Two of the following.
- A) A step up chopper has input voltage of 220V & output voltage of 660V. If the conducting time of thyristor chopper is 100 μ s, compute the pulse width of output voltage. In case output voltage pulse width is halved for a constant frequency operation, Find the average value of new output voltage 6
 - B) Draw and explain 3 phase inverter with 180° conduction mode. Draw phase & line voltages waveforms. 6
 - C) Explain reduction of harmonics in the inverter output voltage. 6
- Q6. Solve Any Two of the following.
- A) Explain the single phase to single phase Circuit- Step up Cycloconverter. 6
 - B) Draw circuit diagram & output voltage and current waveforms for single phase full wave ac voltage controller. 6
 - C) Discuss in details the sequence control of AC voltage controllers. 6

*** End ***